

Hypernatremia — Treatment Calculation

THE CALCULATION APOTHECARY — HYPERNATREMIA: HOW MUCH FREE WATER, HOW SLOW, AND ONGOING LOSSES

Total water needed = baseline deficit + ongoing free water losses — correct ≤ 10 mM/day

2 THE TBW SCALE

9 $TBW = \text{weight} \times \text{factor}$

1	$0.5 \times \text{wt}$
2	0.5×0.5
4	$0.5 / 0.45$

Young ♂ = $0.6 \times \text{wt}$
Young ♀ = $0.5 \times \text{wt}$

8 Young ♂ = $0.6 \times \text{wt}$

7 Elderly ♂ = 0.5 / Elderly ♀ = 0.45

1 THE SLOW-DRIP HOURGLASS

1 Hourglass ≤ 10 mM/day

2 Brain accumulates osmolytes (chronic)

3 Correct deficit over 48–72 h

4 Rapid correction → cerebral edema + seizures

Na > 160 → may need > 48 h

3 THE DEFICIT BEAKER

11 Deficit = $TBW \times (\text{Na} + \text{target} - 1)$

12 Day 1 target = 150, NOT 140

First-day fall ≈ 10 mM only

4 THE WORKED EXAMPLE CHALKBOARD

14 70 kg ♀, Na 160

15 $TBW = 70 \times 0.5 = 35$ L

Deficit Day 1 ≈ 2.3 L (~95 mL/h D5W)

16 70 kg Na 160

6 THE PACING TRAPS TABLET

21 Don't aim for 140 in 24 h

22 Don't forget ongoing urine losses

✓ If overcorrected in adults — do NOT re-induce

5 THE URINE LOSS STATION (EFWC)

17 Ongoing losses — esp. DI

19 $EFWC = V \times (1 - \frac{U_{Na} + U_K}{P_{Na}})$

18 $U_{Na} + U_K$

20 DI patient: e.g. $2.3 + 4.1 = 6.4$ L/day

DI = HUGE

Total water Day 1 = baseline deficit (target 150) + ongoing free-water losses — replace, then re-measure.

1. Total water needed

WHAT YOU SEE

Header: Total water needed = baseline deficit + ongoing free water losses — correct ≤ 10 mM/day

WHAT IT ENCODES

The total daily water prescription has TWO components: the calculated baseline free-water deficit PLUS ongoing free-water losses (urine, insensible ~ 10 mL/kg/day, GI). Correction rate must stay ≤ 10 mEq/L drop in serum Na per 24h. Plan: replace the deficit over 48–72h, add the day's ongoing losses, and recheck Na q4–6h to titrate.

BOARD TRAP

WRONG on both sides: replacing only the deficit and forgetting ongoing losses (undertreats — Na keeps climbing, classic in DI), OR aiming to normalize Na in 24h (overcorrects $\rightarrow$ cerebral edema). Discriminator — deficit + ongoing losses, corrected ≤ 10 mEq/L/day.

2. TBW factor

WHAT YOU SEE

THE TBW SCALE: $0.5/0.6 \times \text{weight by age/sex}$

WHAT IT ENCODES

Total body water (TBW) = weight(kg) $\times$ a fraction that falls with age and female sex due to lower muscle/higher fat: young men 0.6, young women and elderly men 0.5, elderly women 0.45. TBW is the denominator of the deficit formula, so picking the wrong fraction skews the volume to give.

BOARD TRAP

WRONG: using 0.6 for an elderly woman (true factor 0.45). Discriminator — women and the elderly hold proportionally less water (more adipose), so lower fractions; using 0.6 universally overestimates TBW and the deficit.

3. ≤ 10 mM/day

WHAT YOU SEE

Hourglass: 'Correct ≤ 10 mM/day'

WHAT IT ENCODES

The hard safety ceiling: lower serum Na by no more than ~ 10 mEq/L per 24h (some sources say up to 12; ~ 0.5 mEq/L/h). In acute hyponatremia (< 48 h, e.g. salt poisoning) faster correction up to ~ 1 mEq/L/h is tolerated, but for the typical chronic case stay at ≤ 10 /day to protect adapted brain cells.

BOARD TRAP

WRONG: applying a 'fast is fine' rule to chronic hyponatremia. Discriminator — only documented ACUTE hyponatremia tolerates rapid correction; assume chronic and cap at ≤ 10 mEq/L/24h unless onset < 48 h is proven.

4. Osmolytes (chronic)

WHAT YOU SEE

Brain note: brain accumulates osmolytes (chronic)

WHAT IT ENCODES

In chronic hyponatremia, neurons generate idiogenic osmolytes (taurine, myo-inositol, glutamine/glutamate) over ~ 48 h to match serum tonicity and prevent cell shrinkage. These osmolytes clear slowly, so if you drop serum Na faster than the brain can offload them, water rushes into the still-hyperosmolar neurons $\rightarrow$ cerebral edema.

BOARD TRAP

WRONG: assuming the brain re-equilibrates instantly so rapid correction is safe. Discriminator — adaptive osmolytes are why chronic hyponatremia must be corrected slowly; their slow clearance is the mechanism of correction-induced cerebral edema.

5. Correct over 48–72h

WHAT YOU SEE

'Correct deficit over 48–72h'

WHAT IT ENCODES

Replace the total free-water DEFICIT gradually over 48–72 hours, not in a single day. Spreading it out keeps the per-day Na fall within the ≤ 10 mEq/L limit. Combine with frequent Na monitoring (q4–6h) so you can slow or speed the infusion as the trajectory dictates.

BOARD TRAP

WRONG: front-loading the entire deficit on day 1. Discriminator — the deficit is a 48–72h target; day-1 infusion is sized to drop Na ≤ 10 mEq/L, not to erase the whole deficit at once.

6. Rapid $\rightarrow$ edema

WHAT YOU SEE

'Rapid correction $\rightarrow$ cerebral edema + seizures'

WHAT IT ENCODES

Overly rapid lowering of a chronically elevated Na drives free water into osmolyte-loaded neurons $\rightarrow$ cerebral edema, seizures, coma, and death. This is the mirror image of osmotic demyelination in hyponatremia. It is the central safety rule that the whole calculation exists to enforce.

BOARD TRAP

WRONG: blaming new neuro decline during correction on 'worsening hyponatremia' and speeding up fluids. Discriminator — neuro deterioration WHILE the Na is falling fast = iatrogenic cerebral edema; the fix is to SLOW correction, not accelerate it.

7. Free water deficit

WHAT YOU SEE

THE DEFICIT BEAKER: Deficit = TBW $\times$ (Na/target – 1)

WHAT IT ENCODES

Free-water deficit (L) = TBW $\times$ (current Na / target Na – 1), with TBW = weight $\times$ factor. Target Na is ultimately 140, but for day-1 pacing you often calculate toward 145–150 so the planned fall respects ≤ 10 mEq/L. This gives the volume of electrolyte-free water (delivered as D5W or oral water) needed to reach the target.

BOARD TRAP

WRONG: plugging the final target of 140 into the day-1 calculation. Discriminator — day-1 math uses an intermediate target (~ 150) to keep the drop ≤ 10 mEq/L; 140 is the eventual goal over several days, not the first 24h.

8. Day 1 target 150

WHAT YOU SEE

'Day 1: target 150, NOT 140'

WHAT IT ENCODES

For the first day, calculate toward a Na of ~150 (or current Na minus 10), NOT straight to 140. Example: Na 160 → day-1 target 150 = a 10 mEq/L drop, exactly at the ceiling. This staged targeting is how the formula is operationalized safely.

BOARD TRAP

WRONG ANSWER: 'calculate the deficit to reach Na 140 today.' Why wrong — from a starting Na of 160–170 that exceeds the safe 10 mEq/L/24h drop. Discriminator — set day-1 target = starting Na – 10 (≈150), then continue toward 140 on subsequent days.

9. Worked example

WHAT YOU SEE

Chalkboard: 70 kg, Na 160, TBW=70×0.5=35, deficit≈2.3 L, Day1 ~95 mL/h D5W

WHAT IT ENCODES

Worked case: 70 kg patient, Na 160. TBW = 70 × 0.5 = 35 L. Total deficit to 140 = 35 × (160/140 – 1) = 5 L. But for day 1 target 150: 35 × (160/150 – 1) ≈ 2.3 L, given as D5W over 24h ≈ 95 mL/h, PLUS ongoing/insensible losses. Recheck Na to confirm the ~10 mEq/L fall.

BOARD TRAP

WRONG: infusing the full 5 L (the to-140 deficit) on day 1. Discriminator — 5 L is the multi-day deficit; only ~2.3 L (the to-150 portion) plus ongoing losses belongs in day 1 to honor the ≤10 mEq/L cap.

10. EFWC station

WHAT YOU SEE

THE URINE LOSS STATION — EFWC = $V \times (1 - (UNa + UK) / SNa)$

WHAT IT ENCODES

Electrolyte-free water clearance (EFWC) = urine volume × [1 – (UNa + UK)/serum Na] estimates how much pure water the kidney is losing per day. A positive EFWC means ongoing renal free-water loss that must be ADDED to the replacement, or the serum Na will keep rising despite deficit correction.

BOARD TRAP

WRONG: ignoring renal free-water losses and treating only the static deficit. Discriminator — in polyuric states EFWC quantifies the ongoing loss you must replace on top of the deficit; omitting it explains 'why won't the Na come down?'

11. DI = huge losses

WHAT YOU SEE

'DI patient' big jug labeled DI = HUGE losses

WHAT IT ENCODES

In diabetes insipidus, dilute urine output is massive (can exceed 3–10 L/day of nearly free water), so ongoing losses dwarf the baseline deficit. You must replace these losses AND treat the cause — desmopressin (DDAVP) for central DI; remove offending drug / thiazide + low-solute diet for nephrogenic DI — or the Na keeps climbing.

BOARD TRAP

WRONG: calculating a fixed deficit for a DI patient and stopping there. Discriminator — DI is dominated by ENORMOUS ongoing renal free-water losses; without replacing them plus giving DDAVP (central) the Na won't normalize regardless of the deficit math.

12. Don't aim 140 in 24h

WHAT YOU SEE

Tablet: Don't aim for 140 in 24h

WHAT IT ENCODES

Pacing trap #1: do NOT try to normalize Na to 140 within 24h from a high baseline. The per-day fall must stay ≤10 mEq/L, so reaching 140 from 160–170 takes 2–3 days. Set staged daily targets instead of one-shot normalization.

BOARD TRAP

WRONG: prescribing fluids to hit 140 by tomorrow. Discriminator — full normalization in 24h breaches the ≤10 mEq/L/day limit and risks cerebral edema; normalize over 48–72h.

13. Don't forget ongoing

WHAT YOU SEE

Tablet: Don't forget ongoing losses

WHAT IT ENCODES

Pacing trap #2: the calculated deficit is only half the prescription — add ongoing free-water losses (insensible ~10 mL/kg/day, urine via EFWC, GI/diarrhea, fever). Omitting them is the classic reason serum Na fails to fall despite 'correct' deficit replacement.

BOARD TRAP

WRONG: replacing the deficit alone and being puzzled when Na stays high. Discriminator — total day-1 water = deficit + ongoing losses; the missing ongoing-loss term is the usual culprit for stalled correction.

14. Overcorrected → don't re-induce

WHAT YOU SEE

Tablet: If overcorrected in adults, do NOT re-induce

WHAT IT ENCODES

Pacing trap #3: if Na is overcorrected (dropped too fast) in ADULTS, generally do NOT actively re-raise the sodium — just slow/hold free water and monitor, as adults tolerate this better than the reverse. In CHILDREN, where correction-induced cerebral edema risk is higher, relowering/active management may be considered.

BOARD TRAP

WRONG: aggressively re-raising Na with hypertonic saline after an adult overshoots downward. Discriminator — adults: pause free water and observe; the child-specific concern about cerebral edema does not mandate re-inducing hyponatremia in adults.